

PROJECT DOCUMENTATION

Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce

TEAM DETAILS

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Platform	Salesforce Developer Edition
Submission Date	03-10-2026
GitHub	https://github.com/Kazoli11/Customer-Support-Ticket-Agentforce
Demo Video	https://drive.google.com/file/d/15liclAdaE23W9HTXwYe4Z7uylrdhT_3V/view?usp=drivesdk

Naan Mudhalvan Project Documentation

Salesforce Developer Edition

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

User Story:

The Customer Support Ticket Priority Prediction and Automated Assignment System uses Salesforce and Agentforce to analyze customer support ticket descriptions, automatically determine priority as High, Medium, or Low, and trigger backend automation for appropriate support handling. The solution uses the Support Ticket Intelligence custom object, an Auto-Launched Flow, and an Agentforce subagent named Support Ticket Priority Analysis. High-priority tickets can result in an urgent task and senior support assignment, reducing repetitive manual work and supporting faster response to critical issues.

1. INTRODUCTION

1.1 Project Overview :

Customer support teams receive many tickets containing different types of problems. Some requests require immediate attention, while others can be processed through the normal support queue. When prioritization is performed manually, support personnel must read each ticket, interpret the urgency, decide the support level, and then begin the appropriate action.

This project provides a Salesforce-based solution in which ticket information is stored in a structured custom object and analyzed through automation. The system retrieves the latest ticket for a customer account, examines its description, assigns a priority level, and prepares an appropriate action message. Agentforce acts as the conversational interface through which a user can request the analysis.

1.2 Purpose :

- Automatically classify support tickets as High, Medium, or Low priority.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce manual prioritization and assignment effort.
- Create an urgent handling task for High-priority tickets.
- Use Agentforce to request and present the analysis conversationally. Provide a foundation for future SLA monitoring and support-team routing.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	focus on critical tickets first	tickets are not always prioritized automatically	ticket prioritization can be manual	frustrated
PS-2	Customer	get urgent issues resolved quickly	critical issues may wait behind lowerpriority tickets	support teams handle large ticket volumes	concerned
PS-3	Support Manager	assign tickets to the right support level	assignment may require manual decisions	ticket urgency is not consistently analyzed	overwhelmed
PS-4	Support Agent	identify urgent issues from descriptions	important keywords can be missed	ticket descriptions contain different urgency signals	uncertain
PS-5	Support Manager	reduce SLA-related risk	older unresolved tickets may remain unnoticed	SLA risk needs additional checking	concerned

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2.2 Empathy Map Canvas

User Persona: Support Agent

- **Think & Feel:** Wants critical issues identified quickly so urgent customer problems receive attention.
- **Hear:** Receives customer issue descriptions, support requests, and escalation information.
- **See:** Ticket status, issue type, description, priority level, assignment, and SLA-risk information.
- **Say & Do:** Reviews tickets, handles assigned cases, and focuses on urgent support issues.
- **Pain:** Manual prioritization, missed urgency signals, inconsistent assignment, and delays in handling critical tickets.
- **Gain:** Automatic priority classification, clearer assignment, urgent-task creation, and reduced manual effort.

2.3 Brainstorming

The team explored ways to improve support-ticket handling using Salesforce capabilities. The brainstorming focused on combining structured ticket data with automation and conversational interaction. The selected approach uses a custom object for ticket intelligence, Salesforce Flow for deterministic backend processing, and Agentforce for conversational access to the process.

- Use Salesforce as the central platform for customer and ticket data.
- Create a custom Support Ticket Intelligence object.
- Use an Auto-Launched Flow to retrieve and analyze tickets.
- Use Agentforce to collect the Account Name and present results.
- Create an urgent task when a High-priority condition is detected.

3. DESIGN THINKING PHASE

3.1 Customer Journey Map

The customer-support journey begins when a support request is recorded. The support agent or user identifies the customer account, the system retrieves the latest ticket, the description is analyzed, and the resulting priority is returned. High-priority tickets move into urgent handling, while Medium and Low tickets continue through their respective processing paths.

Stage	Customer / Agent Activity	System Response
Ticket Created	Customer issue is recorded.	Ticket information is stored in Salesforce.

Ticket Review	Support user requests analysis.	Agentforce asks for Account Name when required.
Ticket Retrieval	Account is identified.	Flow retrieves the latest matching ticket.
Priority Analysis	Description is evaluated.	Decision logic assigns High, Medium, or Low.
Action	Priority result is processed.	High priority creates urgent task; other priorities return corresponding message.
Follow-up	Support team handles the ticket.	Assignment and priority information guide processing.

3.2 Functional Requirements

- Account-based retrieval of the latest Support Ticket Intelligence record.
- Description-based priority classification.
- High-priority task creation.
- Support-level assignment message.
- SLA breach-risk check based on configured ticket age.
- Agentforce conversational interaction.

3.3 Data Flow Diagram

The main data flow starts with the Account Name supplied through Agentforce. The Flow uses that value to find the Account, retrieves the latest support ticket related to that Account, analyzes the Description field, determines the priority, and returns the resulting variables to the Agentforce action. For High priority, the Flow additionally creates the Urgent Ticket Handling task.

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution Component	Expected Improvement
Manual ticket prioritization	Decision logic in Flow	Consistent High / Medium / Low classification.
Difficulty identifying latest ticket	Account and ticket Get Records steps	Latest relevant ticket is retrieved.
Delayed handling of urgent issues	High-priority task creation	Urgent work is explicitly created.
Manual support-level identification	Assigned To / action message	Support handling level is communicated.
Limited conversational access	Agentforce subagent and action	User can request analysis through conversation.

4.2 Proposed Solution

The proposed solution consists of four connected components. First, Salesforce stores ticket information in the Support Ticket Intelligence object. Second, the Auto-Launched Flow retrieves the relevant customer and latest ticket. Third, the Flow evaluates the ticket description using the configured priority indicators. Fourth, Agentforce provides the conversational entry point and displays the returned result.

4.3 Solution Architecture

The architecture separates data storage, business logic, and conversational interaction. Salesforce data objects provide the source information. The Flow contains the deterministic retrieval and decision logic. Agentforce collects the required input, invokes the action, and presents the returned priority, assignment, ticket ID, account ID, and action message.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project is organized into stages so that the solution can be designed, implemented, tested, and documented systematically. The planning sequence follows the structure of the supplied project template and keeps the Salesforce configuration work connected to the requirements identified during ideation.

Phase	Major Activities	Output
Requirement Analysis	Identify users, problems, and expected behavior.	Problem statement and requirements.
Data Modeling	Create object, fields, and relationships.	Support Ticket Intelligence object.
Automation	Build Auto-Launched Flow.	Ticket retrieval and priority logic.
Agentforce	Create subagent and action.	Conversational ticket analysis.
Testing	Run High, Medium, and Low scenarios.	Validated workflow behavior.
Documentation	Capture implementation and results.	Final project report.

5.2 Sprint / Task Schedule

Task	Description	Status
Data Model	Create custom object and ticket fields.	Completed

Flow	Build ticket retrieval and decision logic.	Completed
High Priority Action	Create urgent task and assignment message.	Completed
Agentforce	Configure subagent, instructions, input, and outputs.	Completed
Testing	Validate priority paths and conversation.	Completed
Documentation	Prepare template-based project report.	Completed

6. PROJECT DEVELOPMENT PHASE

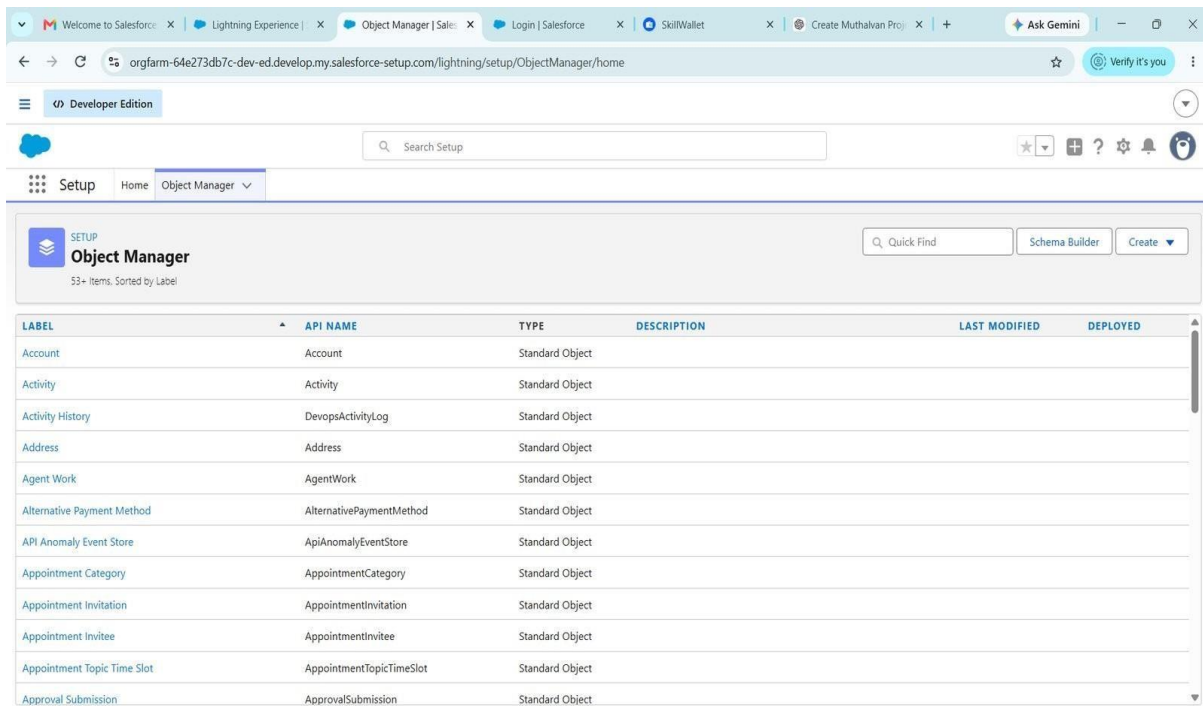
6.1 Data Modelling

The main custom object used in the implementation is **Support Ticket Intelligence**. Its API name is **Support_Ticket_Intelligence__c**. It is designed to store the fields needed for ticket identification, customer association, issue classification, priority, status, assignment, SLArisk awareness, and resolution tracking.

Field	Type / Purpose
Ticket Number	Ticket identifier.
Customer	Lookup to Account.
Contact	Lookup to Contact.
Issue Type	Picklist: Technical, Billing, General.
Description	Long text containing the issue details.
Priority Level	Picklist: Low, Medium, High.
Status	Picklist: New, In Progress, Resolved.
Created Date	Ticket creation timestamp.
Assigned To	Lookup to User.
SLA Breach Risk	Checkbox for SLA-risk indication.
Resolution Time	Number field for resolution duration.

6.2 Fields and Relationships

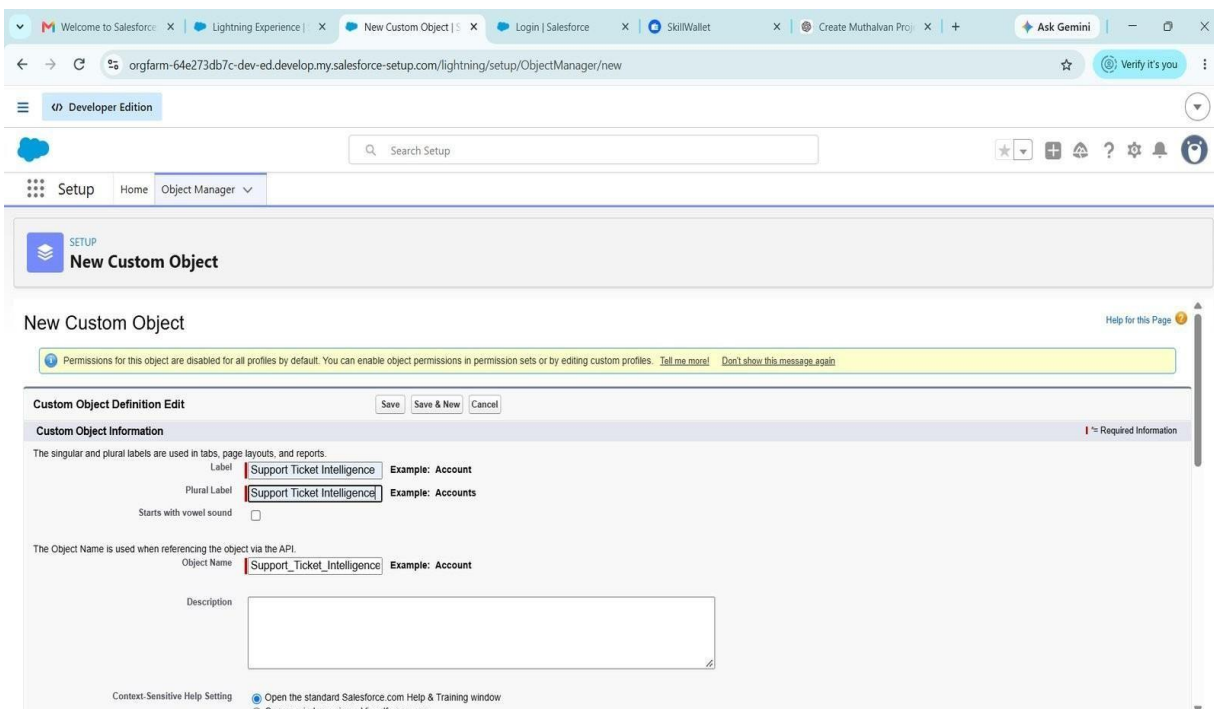
The Customer lookup connects the ticket to the Account. The Contact lookup connects the ticket to the person associated with the request. The Assigned To lookup provides a relationship with the Salesforce User who handles the work. These relationships make the ticket record usable by the Flow during retrieval and assignment.



The screenshot shows the Salesforce Object Manager interface. At the top, there's a search bar and a 'Quick Find' button. Below the search bar, a table lists various standard objects. The table has columns for LABEL, API NAME, TYPE, DESCRIPTION, LAST MODIFIED, and DEPLOYED. The objects listed include Account, Activity, Activity History, Address, Agent Work, Alternative Payment Method, API Anomaly Event Store, Appointment Category, Appointment Invitation, Appointment Invitee, Appointment Topic Time Slot, and Approval Submission.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Account	Account	Standard Object			
Activity	Activity	Standard Object			
Activity History	DevopsActivityLog	Standard Object			
Address	Address	Standard Object			
Agent Work	AgentWork	Standard Object			
Alternative Payment Method	AlternativePaymentMethod	Standard Object			
API Anomaly Event Store	ApiAnomalyEventStore	Standard Object			
Appointment Category	AppointmentCategory	Standard Object			
Appointment Invitation	AppointmentInvitation	Standard Object			
Appointment Invitee	AppointmentInvitee	Standard Object			
Appointment Topic Time Slot	AppointmentTopicTimeSlot	Standard Object			
Approval Submission	ApprovalSubmission	Standard Object			

(a) Object Manager in Salesforce Setup



The screenshot shows the 'New Custom Object' setup page in Salesforce. The page is titled 'New Custom Object' and has a subtitle 'Custom Object Definition Edit'. It includes a 'Save' button, a 'Save & New' button, and a 'Cancel' button. The page is divided into sections for 'Custom Object Information' and 'Custom Object Definition Edit'. The 'Custom Object Information' section includes fields for 'Label' (Support Ticket Intelligence), 'Plural Label' (Support Ticket Intelligence), 'Object Name' (Support_Ticket_Intelligence), and 'Description'. The 'Custom Object Definition Edit' section includes a 'Description' field. The page also has a 'Help for this Page' link and a 'Context Sensitive Help Setting' section.

Custom Object Definition Edit

Save Save & New Cancel

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label **Support Ticket Intelligence** Example: Account

Plural Label **Support Ticket Intelligence** Example: Accounts

Starts with vowel sound ☐

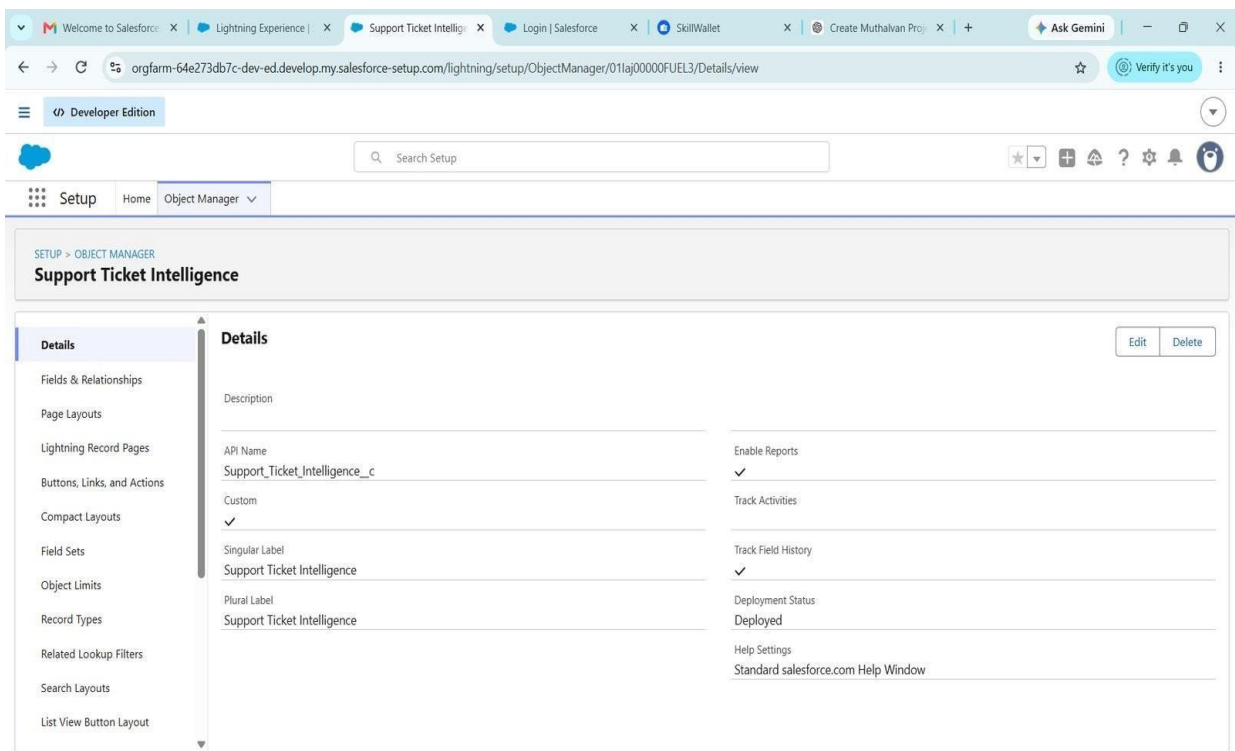
The Object Name is used when referencing the object via the API.

Object Name **Support_Ticket_Intelligence** Example: Account

Description

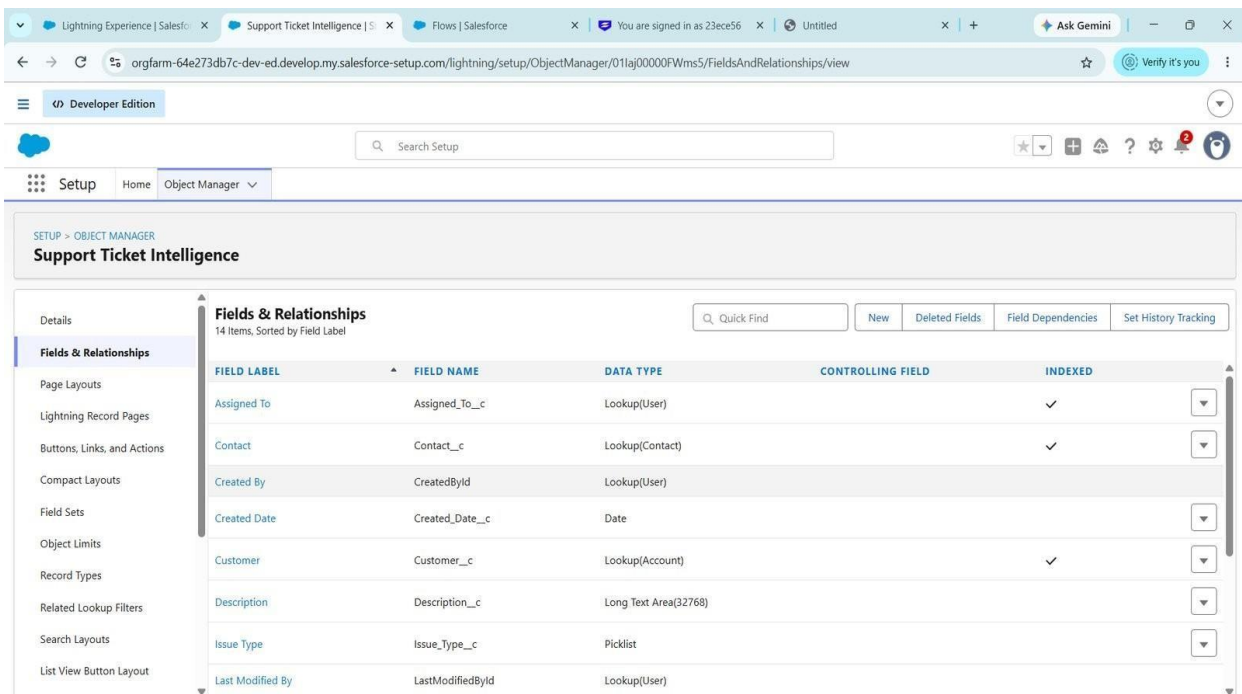
Context Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window ☐ Open a window using a Visualforce page

(b) Creating the new custom object: Support Ticket Intelligence

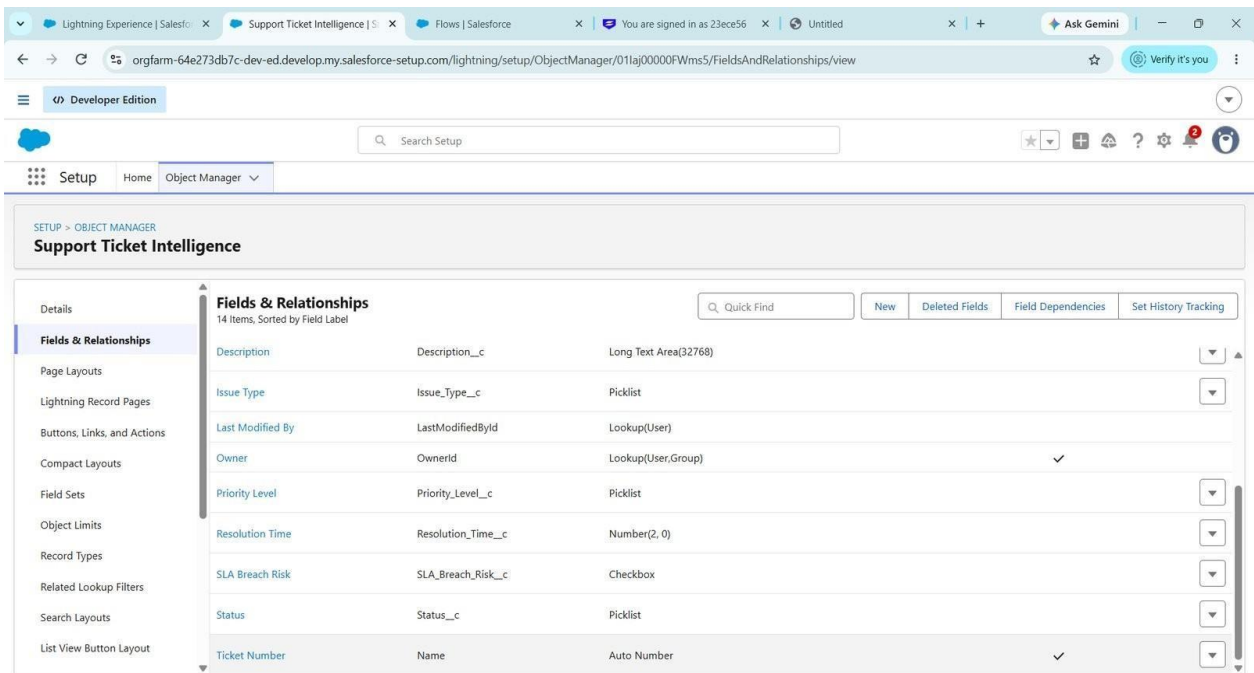


Custom object details with API name Support_Ticket_Intelligence__c

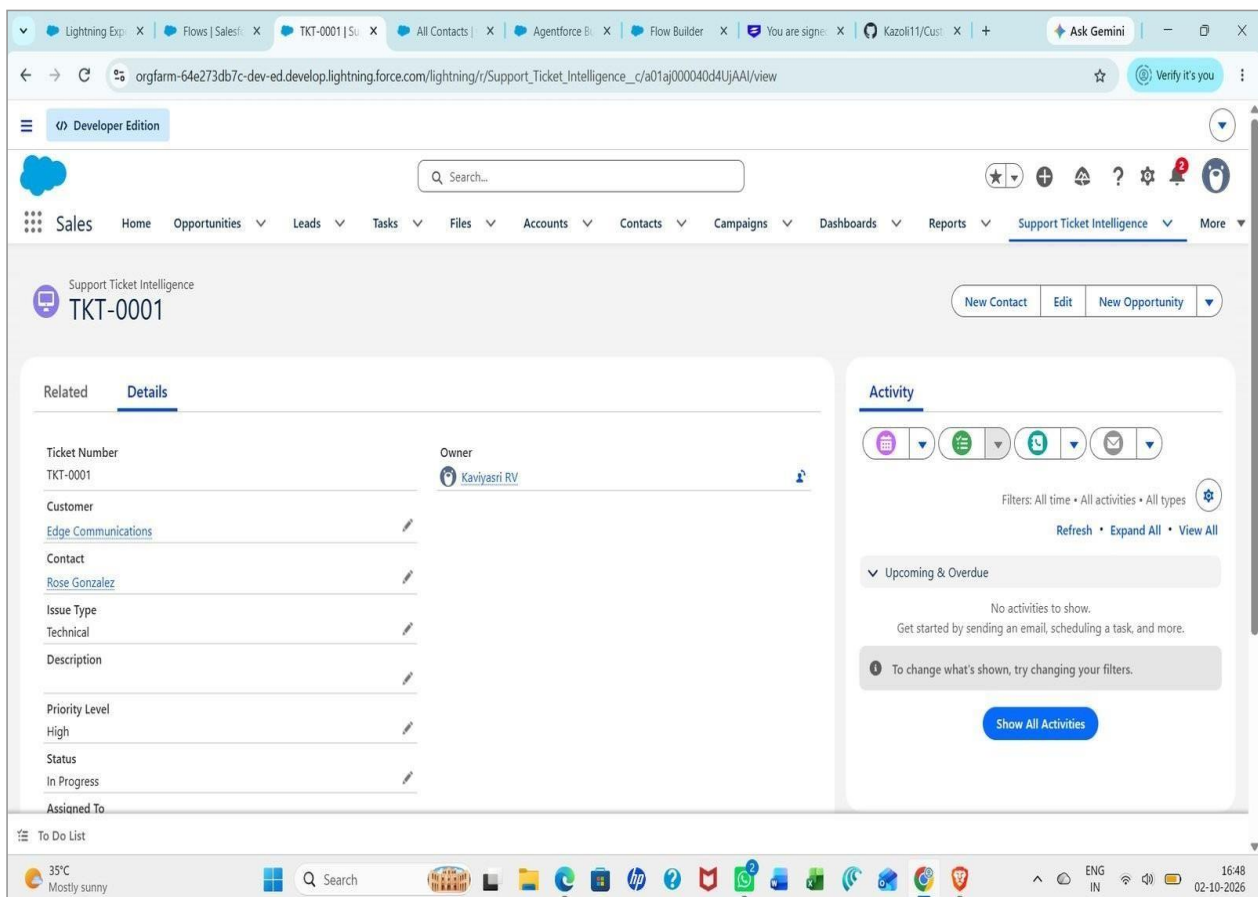
(c)



(d) Fields & Relationships of the object (1 of 2)



(e) Fields & Relationships of the object (2 of 2)



(f) Sample ticket record TKT-0001 stored in the object

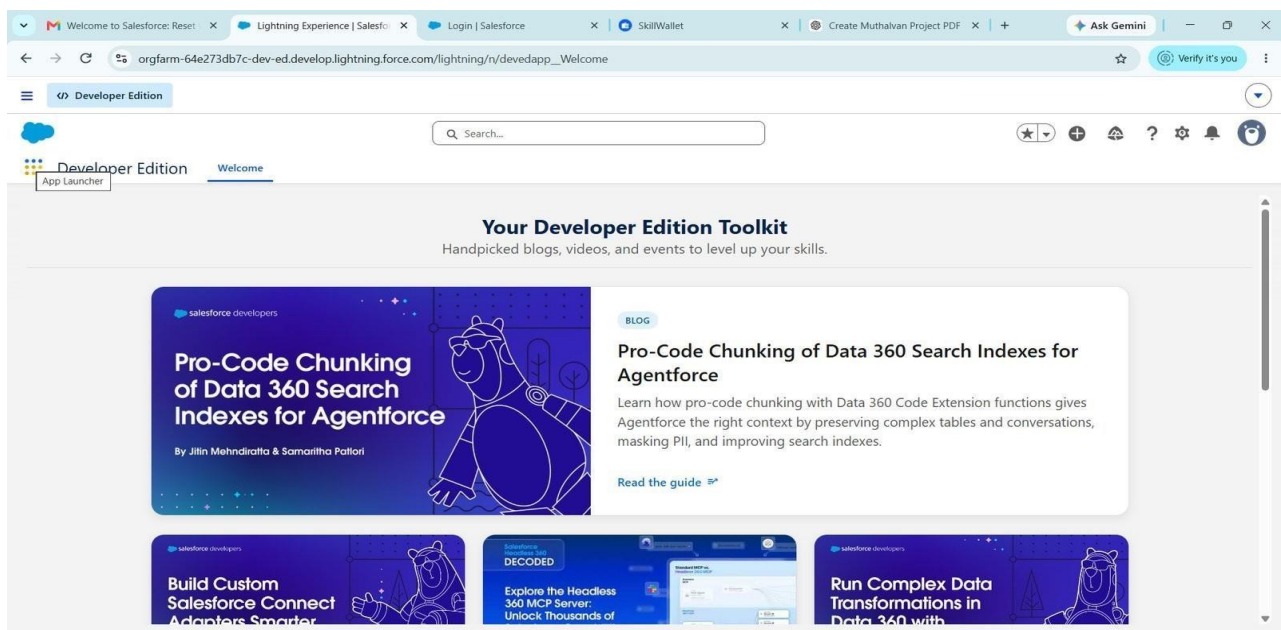
Figure 3: Support Ticket Intelligence custom object and fields

6.3 Salesforce Setup and Developer Environment

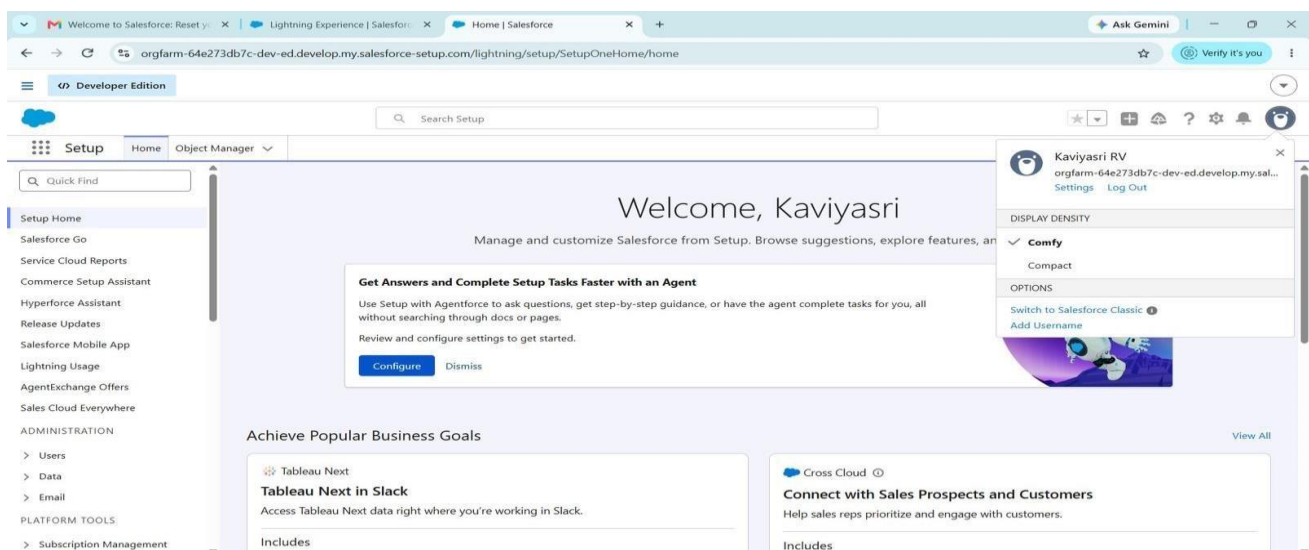
The Salesforce Developer Edition environment is used for the project. The setup process includes creating or accessing the developer org, opening Salesforce Setup, configuring the required object, and preparing the Flow and Agentforce components.

6.4 Auto-Launched Flow Logic

The Flow is designed as an Auto-Launched Flow because it is intended to be invoked by the Agentforce action rather than started directly by a record-trigger event. The Flow receives the Account Name and returns the information required by the conversational layer.



(a) Salesforce Developer Edition org home page



(b) Salesforce Setup home page

Figure 4: Salesforce Developer Edition setup / Setup page

6.5 Flow Type

The selected Flow type is Auto-Launched Flow. It accepts input from Agentforce and executes the ticket analysis sequence without requiring a screen-based user interaction. **6.6**

Variables

Variable	Type / Role	Input / Output
varAccountName	Text; customer Account Name.	Input
varAccountId	Account record ID.	Output
varTicketId	Latest ticket record ID.	Output
varPriorityLevel	Calculated priority.	Output
varAssignedTo	Support assignment information.	Output
varActionMessage	Final user-facing message.	Output

6.7 Step-by-Step Flow Implementation

Step 1: Get Account – Search the Account object using the supplied Account Name. Records are ordered by Created Date in descending order and the first matching record is selected.

Step 2: Store Account ID – Save the selected Account ID for use in the ticket lookup.

Step 3: Get Ticket Record – Search Support_Ticket_Intelligence__c for the latest ticket whose Customer field matches the Account ID.

Step 4: Store Ticket ID – Store the retrieved ticket ID for subsequent task creation and output.

Step 5: Decision – Analyze Description – Compare the ticket Description against the configured High and Medium keyword groups.

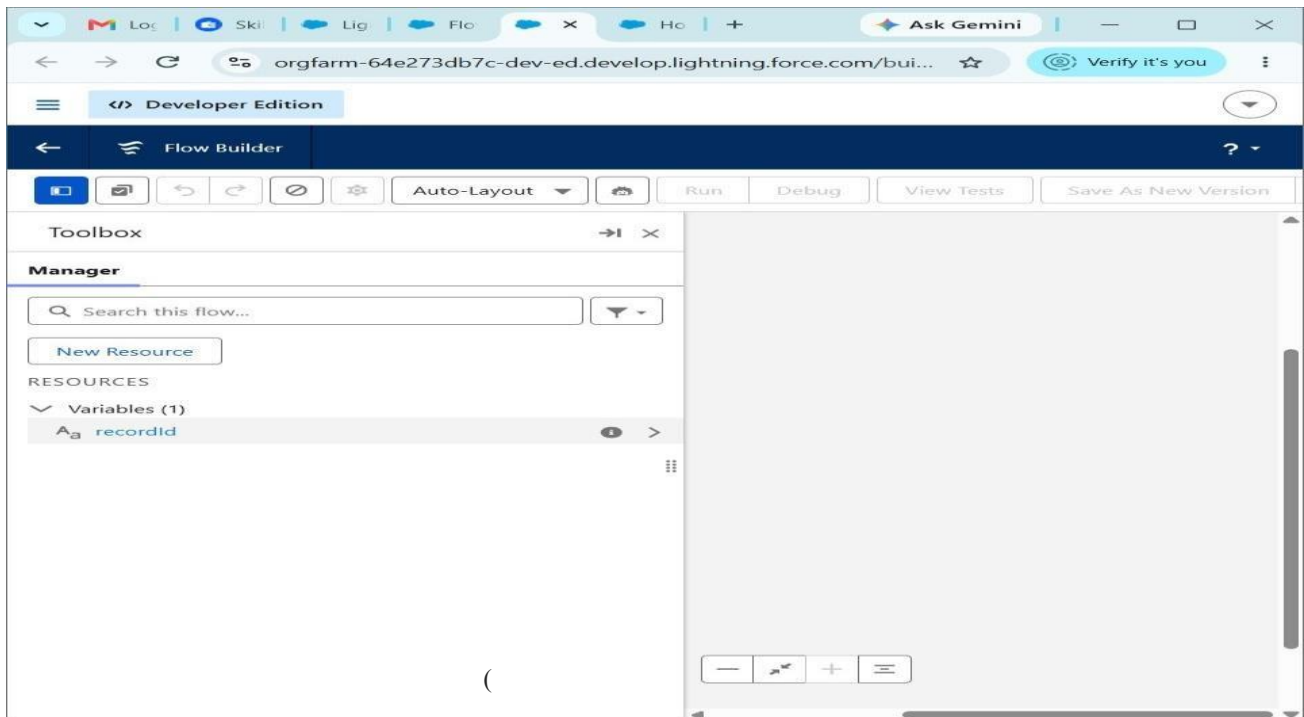
Step 6: Set Priority – Store High, Medium, or Low in varPriorityLevel.

Step 7: Decision – High Priority Check – Determine whether the ticket follows the urgent handling path.

Step 8: Create Task – For High priority, create a Task with Subject 'Urgent Ticket Handling', WhatId equal to the ticket, Priority High, and Status Not Started.

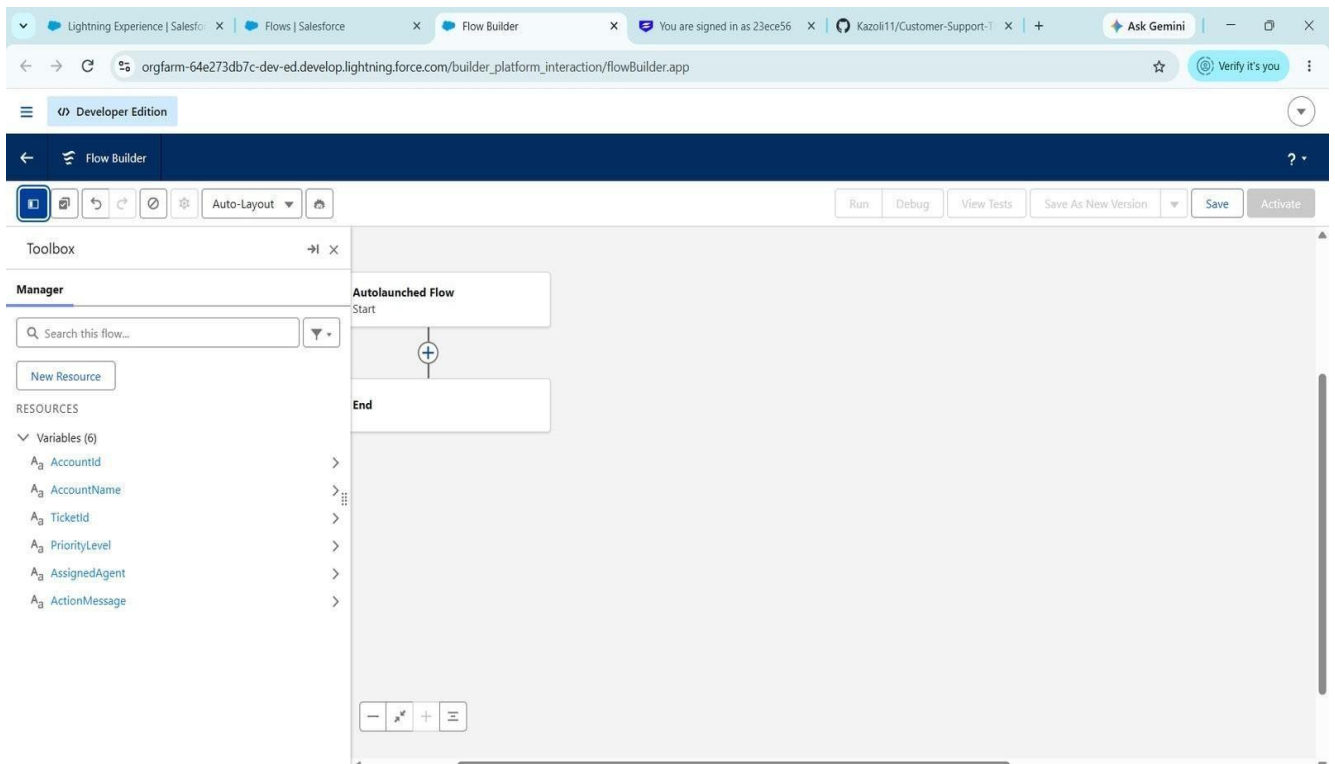
Step 9: Assignment – Set the assignment result to the configured Senior Support Agent level.

Step 10: Optional SLA Check – Identify tickets whose Created Date is older than two days.

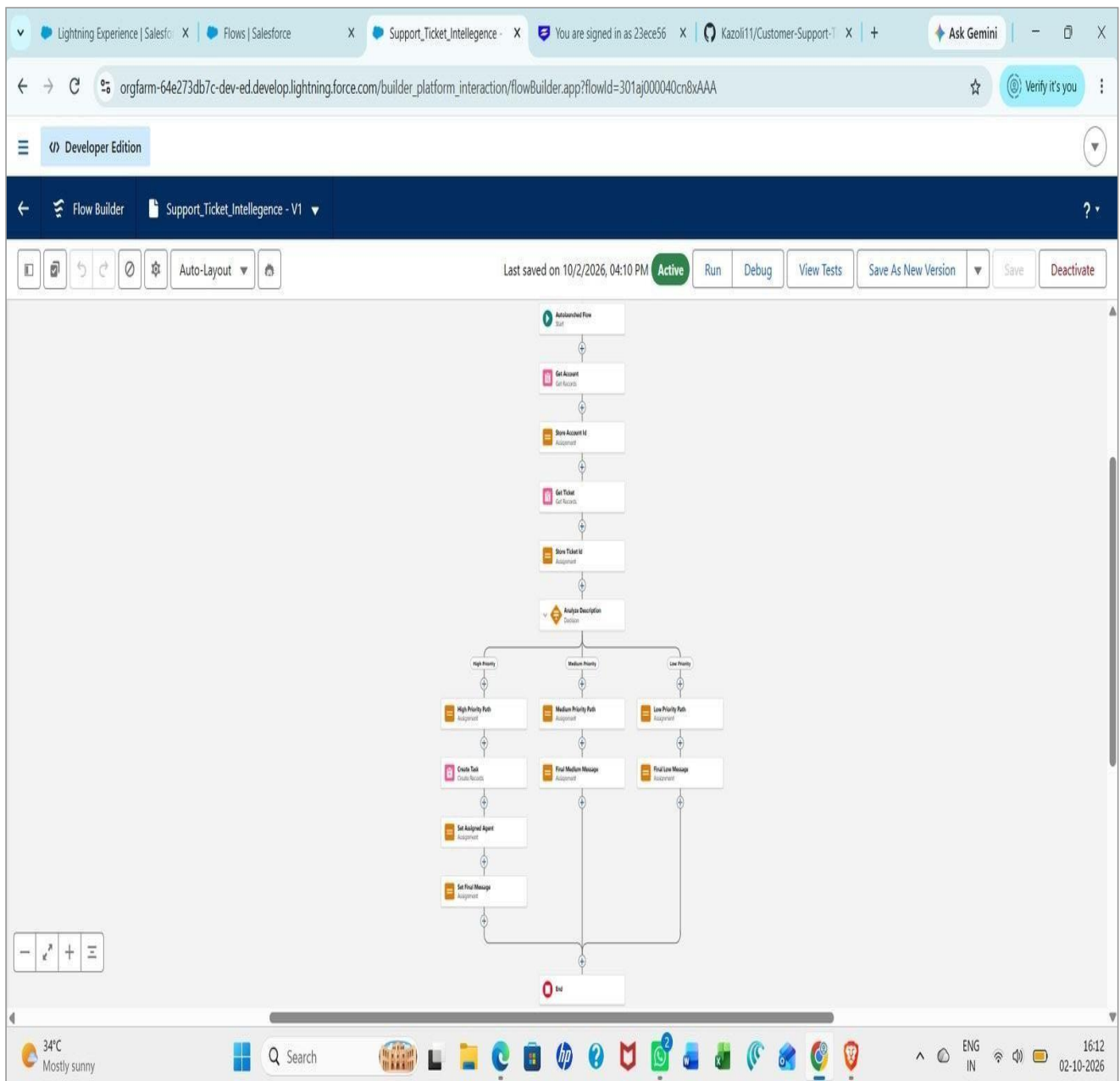


a) New Auto-Launched Flow opened in Flow Builder

Step 12: Save Flow – Save the completed Flow for use by the Agentforce action.



(c) Flow resources: input and output variables



(d) Completed and activated flow (Support_Ticket_Intelligence)

Figure 5: Auto-Launched Flow Builder showing the complete implementation

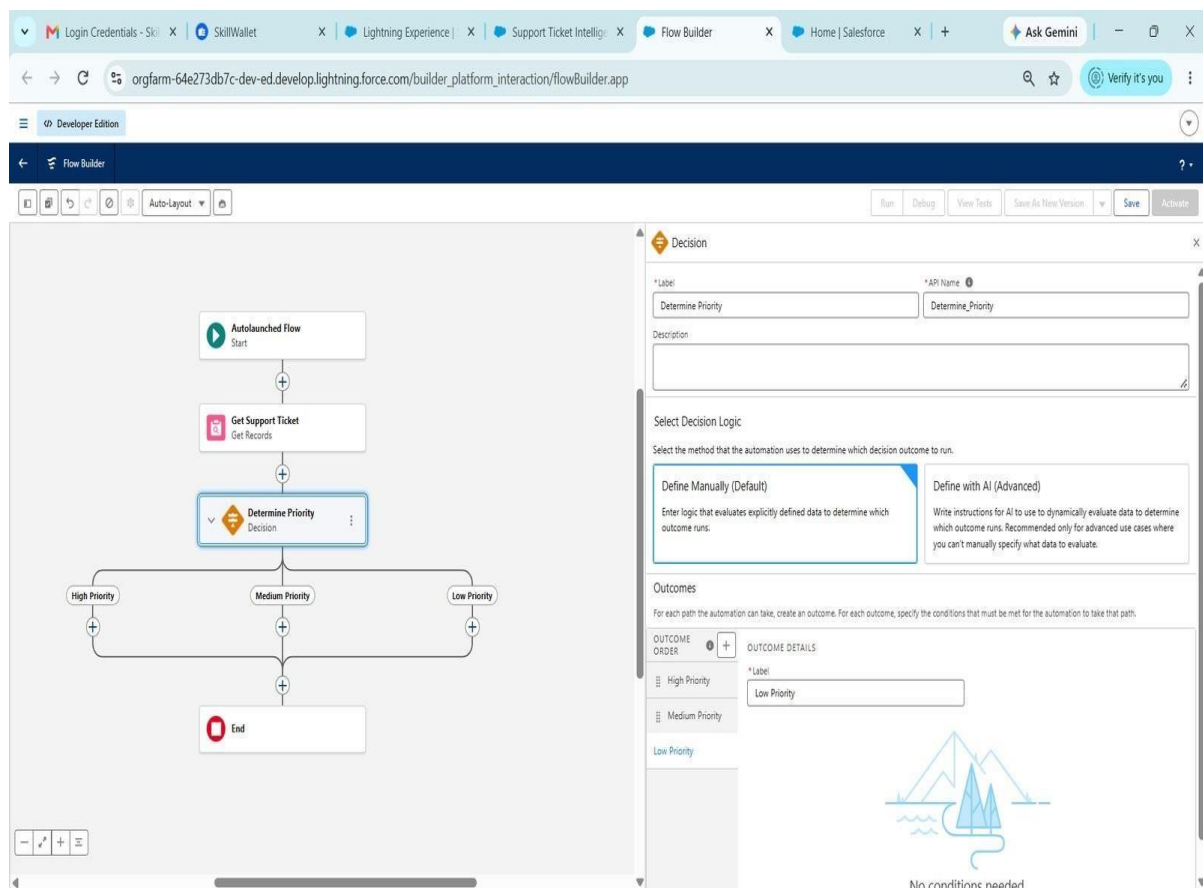
6.8 Priority Decision Rules

Priority	Description Condition	Action
High	Contains urgent, not working, or failure.	High priority; urgent task; senior support handling.

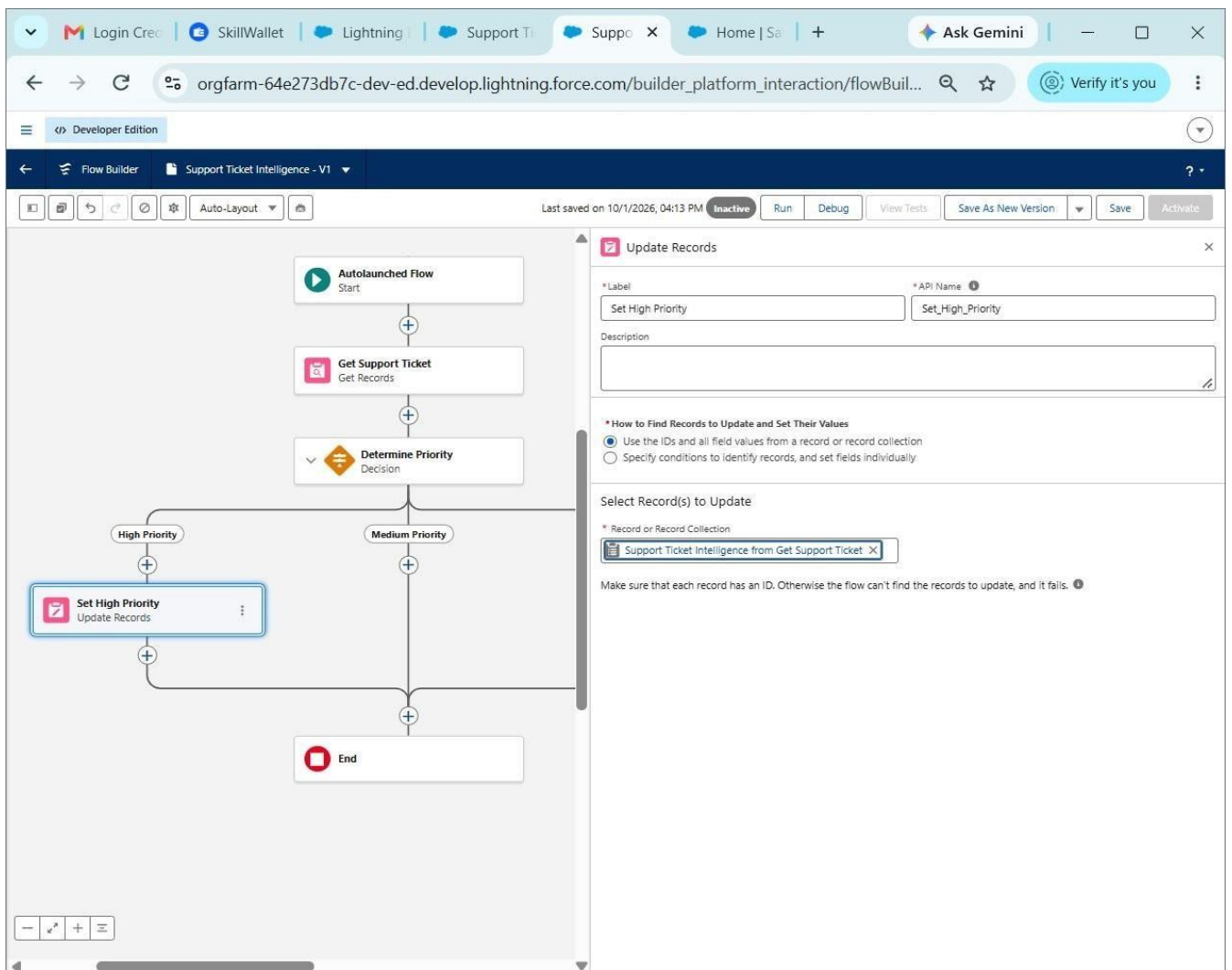
Medium	Contains issue, slow, or delay.	Medium priority; normal short-term handling message.
Low	No configured High or Medium indicator.	Low priority; queued for processing.

6.9 Final Outcome

The Flow returns a set of outputs to the Agentforce action. These outputs allow the agent to identify the Account, ticket, calculated priority, assignment information, and final action message. This provides a complete response without requiring the user to inspect the backend Flow.



(a) Determine Priority decision with High, Medium and Low outcomes



(b) Update Records element configured for the High Priority branch

(c) Figure 6: Flow structure, priority decision branches, and final output

6.10 Agentforce Subagent Configuration

A dedicated Agentforce subagent is configured with the name **Support Ticket Priority Analysis** and API name **Support_Ticket_Priority_Analysis**. The classification description defines the purpose of the subagent as analyzing customer support tickets to determine priority from issue description and urgency.

6.11 Scope

The subagent is limited to ticket-description analysis, High/Medium/Low priority determination, and triggering the configured backend automation. It is not intended to perform billing, subscription, or account-update operations.

6.12 Instructions

- Ask for Account Name.
- Retrieve the latest ticket associated with the account.
- Analyze the description using the defined keyword rules.

- Trigger the backend Flow for High-priority handling.
- Use the Flow for record actions rather than directly updating records.
- Return a clear result containing the relevant priority and action information.

6.13 Agent Action Description

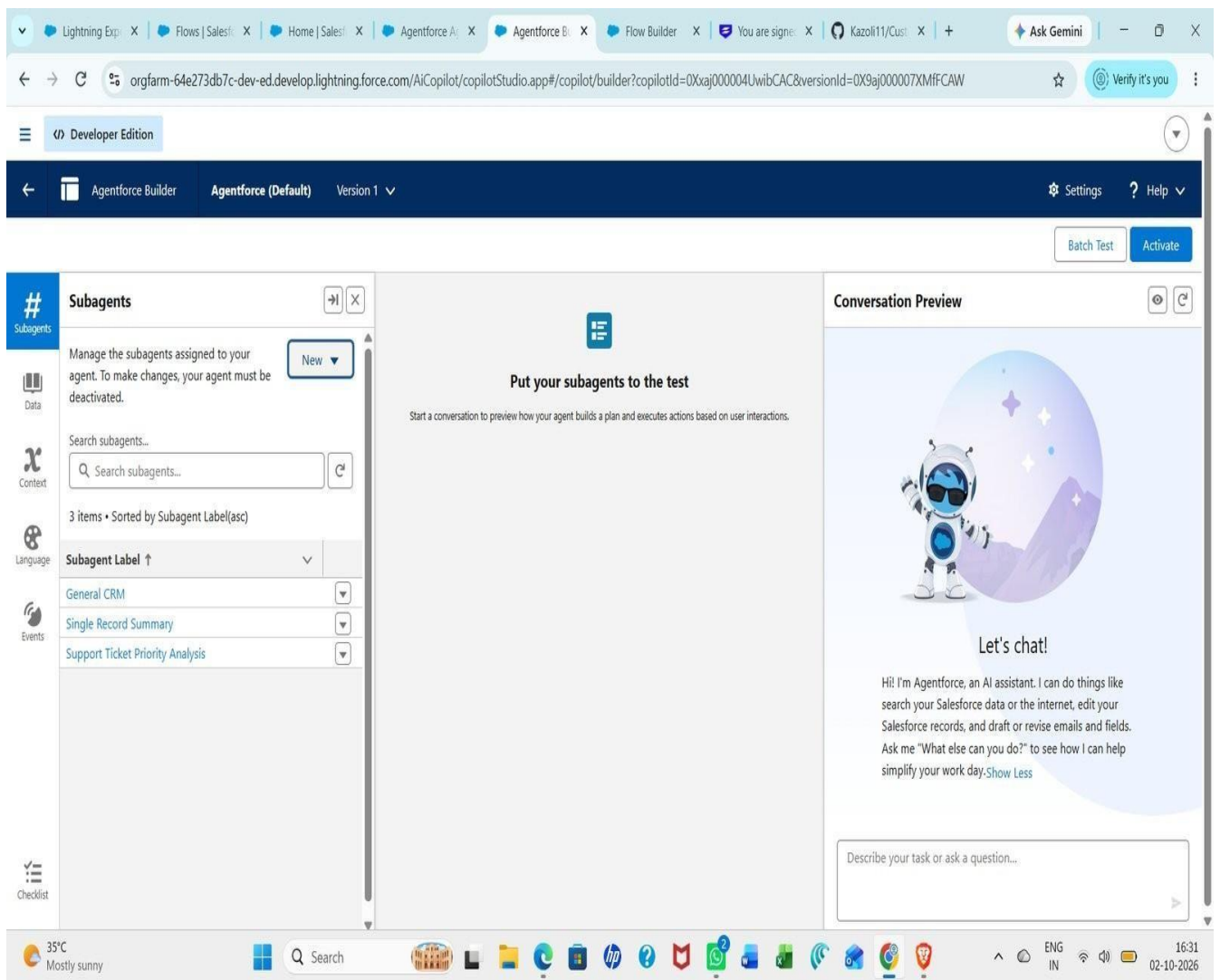
The Agentforce action is configured to analyze a support ticket using Account Name, determine its priority, and trigger automation for High-priority cases. The input variable is **varAccountName**, configured as required text input with data collection enabled.

The screenshot shows the 'Create a Subagent' form in the Agentforce Builder interface. The form is titled 'Create a Subagent' and has a close button in the top right corner. The form is divided into several sections:

- Name:** Support Ticket Priority Analysis
- API Name:** Support_Ticket_Priority_Analysis
- Classification Description:** This topic analyzes customer support tickets to determine priority based on issue description and urgency. It handles user requests related to ticket status, issue severity, and support prioritization.
- Scope:** Your job is only to analyze support ticket descriptions, determine the priority level (High, Medium, Low), and trigger backend automation for task assignment. Do not handle unrelated requests such as billing, subscription management, or account updates.
- Instructions:** The following instructions are used to run this subagent.
 - Instruction:** Ask the user to provide the Account Name if not already given.

At the bottom of the form, there is a progress bar with a blue circle indicating the current step. To the left of the progress bar is a 'Back' button, and to the right is a 'Next' button.

(a) Create a Subagent: name, API name, classification description, scope and instructions



(b) Support Ticket Priority Analysis listed among the agent's subagents

Figure 7: Agentforce subagent configuration, classification description, and scope

6.14 Loading Text and Input

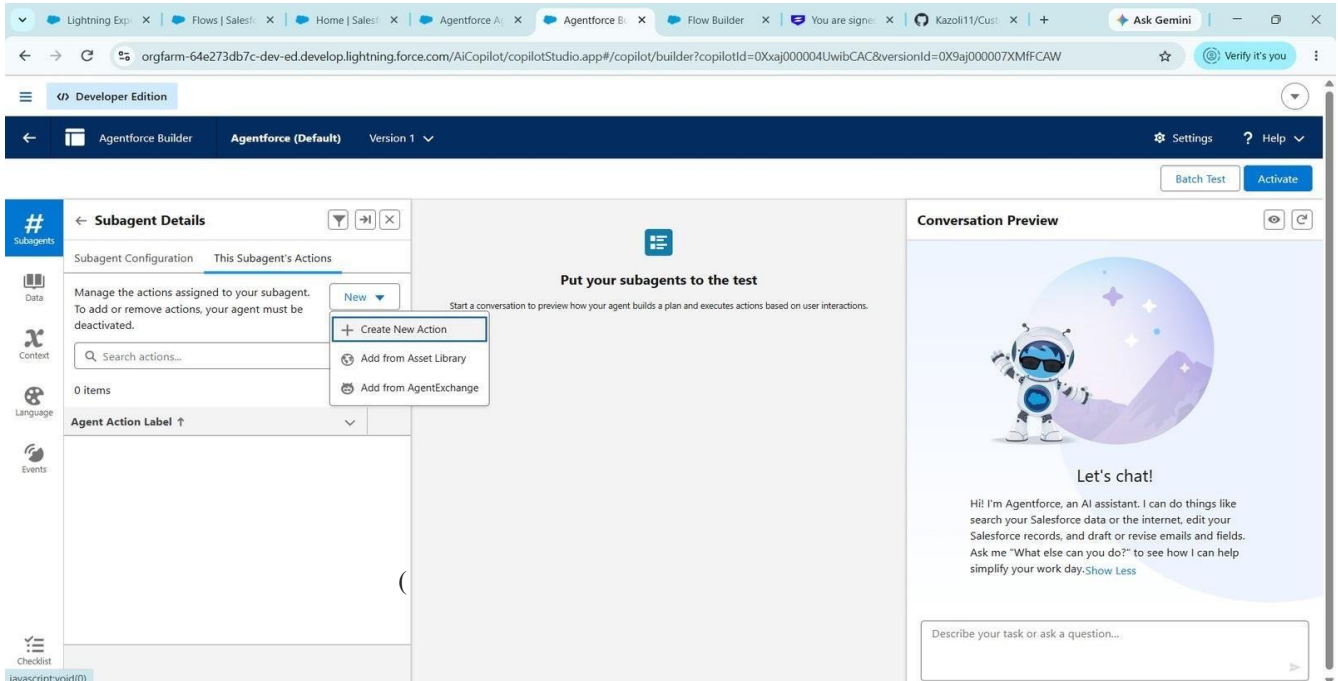
The agent action uses the Account Name as the main input. The action can collect this value during the conversation and pass it into the Flow. This provides a simple conversational starting point while keeping the actual ticket lookup in the backend automation.

6.15 Outputs

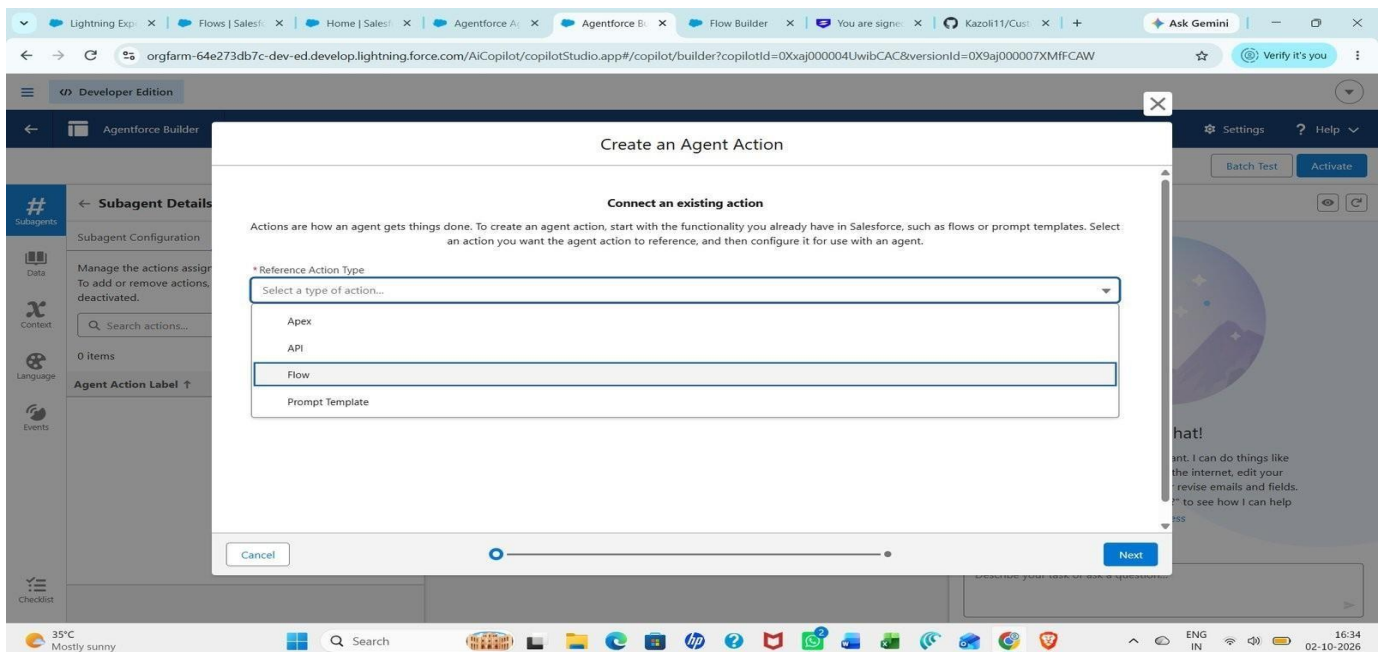
Output	Purpose
varAccountId	Identifies the Account used in the analysis.
varActionMessage	Returns the final priority/action response.
varAssignedTo	Returns the configured support assignment.
varPriorityLevel	Returns High, Medium, or Low.
varTicketId	Identifies the analyzed support ticket.

6.16 Conversation Preview

The completed Agentforce interaction should collect the Account Name, invoke the supportticket priority action, and present the returned result in a conversational format. The conversation therefore acts as the user-facing layer, while Salesforce Flow performs the deterministic backend processing.



a) Subagent's Actions tab: New > Create New Action



(b) Selecting Flow as the reference action type

Create an Agent Action

Connect an existing action

Actions are how an agent gets things done. To create an agent action, start with the functionality you already have in Salesforce, such as flows or prompt templates. Select an action you want the agent action to reference, and then configure it for use with an agent.

*Reference Action Type
Flow

*Reference Action
Support_Ticket_Intelligence

*Agent Action Label
Support_Ticket_Intelligence

*Agent Action API Name
Support_Ticket_Intelligence

Cancel Next

(b) Referencing the Support_Ticket_Intelligence flow

Create an Agent Action

Configure your action for Agent

An agent uses a large language model (LLM) to determine when to launch an action in a conversation. Enter instructions to tell the LLM how to use your action, inputs, and outputs. To help you get started, we copied over details from the reference action. [Learn more about how to write agent action instructions in Salesforce Help.](#)

After you create your action, you can add it to an agent and test and iterate on your instructions.

Agent Action Configuration

Agent Action Label
Support_Ticket_Intelligence

*Agent Action Description
Analyzes customer support tickets based on the provided account name, determines ticket priority using issue description, and triggers automation to assign tasks for high-priority cases.

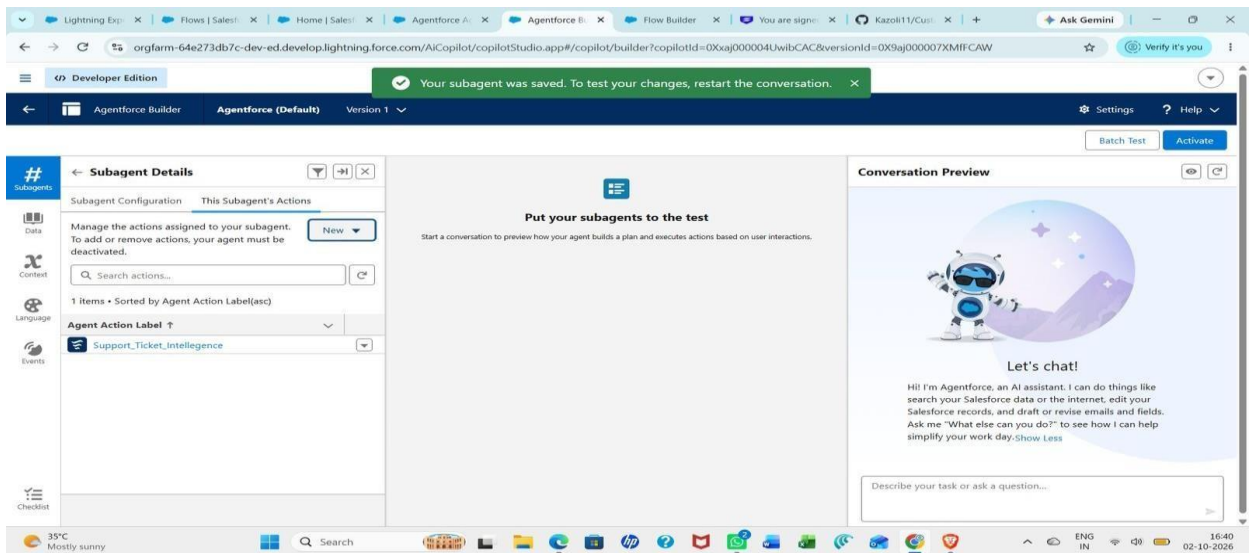
Show loading text for this action
☒

*Loading Text
Analyzing ticket details and determining priority...

Inputs Outputs

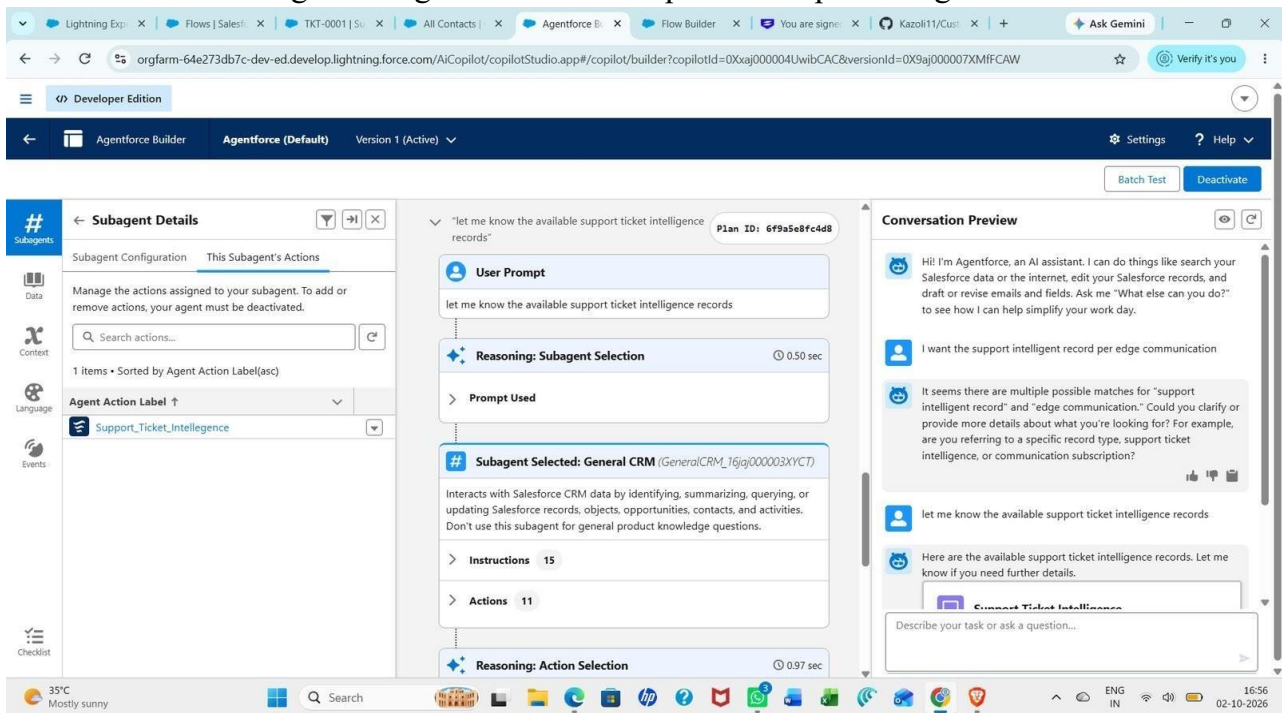
Back Finish

(c) Agent action description and loading text



(e) Action saved and listed under the subagent

Figure 8: Agentforce action input and output configuration



(a) Agentforce conversation preview with the Support_Ticket_Intelligence action

Figure 9: Agentforce conversation preview and returned priority result

7. FUNCTIONAL & PERFORMANCE TESTING

7.1 Testing Approach

Testing is performed by using representative ticket descriptions and checking whether the expected priority branch and backend action are executed. The main validation scenarios are

High, Medium, and Low priority. The Agentforce conversation is also checked to ensure that the input and returned outputs are connected correctly.

7.2 Functional Test Cases

Test ID	Scenario	Expected Result
FT-01	Description contains 'urgent'.	High priority and urgent task creation.
FT-02	Description contains 'not working'.	High priority and urgent handling.
FT-03	Description contains 'failure'.	High priority and urgent handling.
FT-04	Description contains 'issue'.	Medium priority response.
FT-05	Description contains 'slow' or 'delay'.	Medium priority response.
FT-06	No configured priority keyword.	Low priority response.
FT-07	Valid Account Name supplied through Agentforce.	Latest matching ticket is retrieved.
FT-08	High-priority result returned.	Priority, assignment, ticket ID, and action message are available.

7.3 Performance and Reliability Considerations

The workflow uses a limited sequence of record retrieval and decision steps. The design reduces unnecessary user interaction by collecting only the Account Name before the backend

Flow performs the analysis. Reliability depends on the presence of correct Account and ticket data, correct Flow configuration, and correct Agentforce action mapping.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Automatic ticket prioritization.
- Faster identification of critical support issues.
- Reduced repetitive manual classification.
- Automatic urgent-task creation for High priority.
- Clear support-level assignment.
- Conversational access through Agentforce.
- SLA-risk awareness through the configured age condition.
- Centralized Salesforce implementation.
- Extensible design for additional support rules.

8.2 Disadvantages / Limitations

- Keyword-based logic may not understand every context or variation of language.
- The result depends on accurate ticket data.
- Changes to business rules require Flow maintenance.
- Agentforce behavior depends on correct instructions and action configuration.
- Current assignment logic is limited compared with a multi-team routing system.
- SLA monitoring is currently limited to the configured condition.
- The current design has limited analytics compared with a full support dashboard solution.

9. CONCLUSION

The Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce demonstrates a structured approach to automating customer-support ticket handling in Salesforce. The custom Support Ticket Intelligence object stores the required information, while the Auto-Launched Flow retrieves the latest ticket and applies the configured priority decision rules.

Agentforce provides the conversational interface for requesting the analysis. The action passes the Account Name to the Flow and receives the calculated priority, ticket

information, assignment result, and final action message. For High-priority tickets, the Flow creates an urgent handling task and identifies the Senior Support Agent level.

The completed implementation brings together data modeling, Salesforce automation, Agentforce configuration, and functional testing. The design also provides a base for future enhancements such as richer priority indicators, advanced SLA escalation, multiple support teams, and analytics.

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10. FUTURE SCOPE

10.1 Enhanced Priority Analysis

Additional priority indicators and combinations can be added to cover a wider range of support situations. Future rules could consider more than a single keyword and could incorporate additional ticket attributes.

10.2 Advanced SLA Monitoring

The solution can be extended to support multiple SLA thresholds, automated escalation, reminders, and different rules for different issue categories.

10.3 Multi-Team Assignment

Future versions can route tickets to different support teams according to issue type, priority, customer category, or specialist requirement.

10.4 Analytics and Reporting

Dashboards and reports can be added to show ticket volumes, priority distribution, response time, resolution time, SLA-risk records, and assignment patterns.

10.5 Richer Agentforce Interaction

Agentforce can be extended to provide more detailed explanations and guided support workflows while continuing to use Salesforce Flow for controlled backend actions.

11. APPENDIX

11.1 Project Information

Project Title: Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce

Team ID: SWTID-2026-5343

Team Lead: Kaviyasri RV

Team Members: Vijayadharshini B, Kiruthika S, Deepakala C M, Joshikha J

Platform: Salesforce Developer Edition

Submission Date: 02-10-2026

GitHub: To be added

Demo Video: To be added

11.2 Key Configuration Names

Custom Object: Support Ticket Intelligence

API Name: Support_Ticket_Intelligence__c

Flow: Support_Ticket_Intelligence

Agentforce Subagent: Support Ticket Priority Analysis

Agent API Name: Support_Ticket_Priority_Analysis

Input: varAccountName

Outputs: varAccountId, varActionMessage, varAssignedTo, varPriorityLevel, varTicketId